

Completing Parametric Unimodular Rows to Unimodular Matrices

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Serre, in his 1955 paper [5], remarked that the corresponding question was not known for algebraic vector bundles: "It is not known whether there exist projective A -modules of finite type which are not free." Here A is a polynomial ring over a field. Serre made some progress towards a solution in 1957 when he proved that every finitely generated projective module P over a polynomial ring over a field was *stably free*, i.e., $P \oplus A^t \simeq A^s$. In view of this, Serre's Problem becomes the following: does "stably free" imply "free" over $k[x_1, \dots, x_n]$? In fact, in the aspect of polynomial matrix, this problem is equivalent to the following problem (refer to [7], Proposition 20):

If $f_1, \dots, f_m \in A = k[x_1, \dots, x_n]$ generate the unit ideal, can we always complete $(f_1, \dots, f_m)$ to an $m \times m$ matrix over A with determinant in $k \setminus \{0\}$?

In 1976, Quillen [3] and Suslin [6] proved independently that Serre's Conjecture is true and gave a positive answer to the problem above, which means that polynomial vector $(f_1, \dots, f_m)$ can be completed to a unimodular matrix if $f_1, \dots, f_m$ generate the unit ideal. This theorem is commonly referred to Quillen-Suslin theorem:

Quillen-Suslin Theorem. *A polynomial vector $(f_1, \dots, f_m)$ can be completed to a unimodular matrix if $f_1, \dots, f_m$ generate the unit ideal.*

The constructive version of the Quillen-Suslin theorem is presented by [4] for $A = k[x_1, x_2]$ and by [2] for $A = \mathbb{C}[x_1, \dots, x_n]$. The first implementation of the constructive versions of Quillen-Suslin algorithm in computer algebra system *Maple* is presented in [1] for $A = \mathbb{Q}[x_1, \dots, x_n]$.

In this paper, we will extend the Quillen-Suslin Theorem to parametric case. Specifically, we will consider the following problem: Let L be an algebraic closed field contained in k , $X = (x_1, \dots, x_n)$ and $U = (u_1, \dots, u_s)$.

Problem 1. Let $\vec{f}(U, X) \in k[U][X]^{1 \times m}$ be a polynomial vector with variables $x_1, \dots, x_n$ and parameters $u_1, \dots, u_s$. Find a series of subsets $C_1, \dots, C_s$ of L^s and polynomial matrices $P_1, \dots, P_s \in k[U][X]^{m \times m}$, such that $C_1 \cup \dots \cup C_s = L^s$ and for any $(a_1, \dots, a_s) \in C_i$, we have

1. $\vec{f}(a_1, \dots, a_s, X) \in L[X]^{1 \times m}$ is completed to $P_i(a_1, \dots, a_s, X) \in L[X]^{m \times m}$, i.e., $\vec{f}(a_1, \dots, a_s, X)$ is the first row of $P_i(a_1, \dots, a_s, X)$.
2. $P_i(a_1, \dots, a_s, X)$ is unimodular, i.e., $\det(P_i(a_1, \dots, a_s, X)) \in L$.

Example 2. Let

$$\begin{aligned}\vec{f}(u, x, y) &= (-uy - y, x^2 + ux + y^2 + xy + x, x^2 - ux - y, y^2 + uy - u^2 + u) \\ &= (f_1, f_2, f_3, f_4, f_5) \in \mathbb{Q}[u][x, y]^{1 \times 5}.\end{aligned}$$

By computation, we complete $\vec{f}$ to unimodular matrices in parametric case and obtain the following result: For any $a \in \mathbb{C}$, we have

- (1) if $(a - 1)a = 0$, then $\vec{f}(a, x, y)$ can not be completed to a unimodular matrix;
- (2) if $(a - 1)a \neq 0$ and $(a + 1)(2a + 1) \neq 0$, then $\vec{f}(a, x, y)$ can be completed to a unimodular matrix P_1 over $\mathbb{C}[x, y]$, where

$$P_1 = \begin{pmatrix} -ay - y & x^2 + ax + y^2 + xy + x & x^2 - ax - y & y^2 + ay - a^2 + a \\ a + 1 & -y - x & 1 & -y - a \\ 0 & -x & 2a - x + 1 & 0 \\ 0 & -a - x - 1 & a - x & 0 \end{pmatrix};$$

- (3) if $a + 1 = 0$, then $\vec{f}(a, x, y)$ can be completed to a unimodular matrix P_2 over $\mathbb{C}[x, y]$, where

$$P_2 = \begin{pmatrix} 0 & x^2 + y^2 + xy & x^2 + x - y & y^2 - y - 2 \\ 0 & x^2 + 2x + y & -1 & x^2 + x + y - 1 \\ 1 & 0 & 0 & 0 \\ 0 & x^2 - 4xy + y^2 + x - 3y - 14 & x^2 + 6x - y + 3 & -5xy + y^2 - 4x - 4y - 10 \end{pmatrix};$$

- (4) if $2a + 1 = 0$, then $\vec{f}(a, x, y)$ can be completed to a unimodular matrix P_3 over $\mathbb{C}[x, y]$, where

$$P_3 = \begin{pmatrix} -\frac{y}{2} & x^2 + xy + y^2 + \frac{1}{2}x & x^2 + \frac{1}{2}x - y & y^2 - \frac{1}{2}y - \frac{3}{4} \\ -1 & 2x + 2y & -2 & -1 + 2y \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$

Notations 3. For $\vec{a} = (a_1, \dots, a_s) \in L^s$ and $\vec{f} \in k[U][X]^{1 \times m}$, we denote the vector obtained by substituting $u_1, \dots, u_s$ with $a_1, \dots, a_s$ as $\vec{f}(\vec{a})$.

In the following, we sketch our idea for solving Problem 1:

- (1) By computing the comprehensive Gröbner system of $\{f_1, \dots, f_m\}$, we can find a constructible set $C \subset L^s$ such that for any $\vec{a} = (a_1, \dots, a_s) \in L$, $\vec{f}(\vec{a})$ is unimodular if and only if $\vec{a} \in C$.
- (2) Without loss of generality, suppose that $C = V(E) \setminus V(N)$, where $V(E)$ and $V(N)$ are the affine varieties defined by ideal $E, N \subset k[U]$ respectively. Compute the irreducible decomposition of $V(E)$, such that $V(E) = \cup_{i=1}^l V(E_i)$ with each E_i is a prime ideal in $k[U]$. Then $C = \cup_{i=1}^l (V(E_i) \setminus V(N))$.

- (3) For each $C_i = V(E_i) \setminus V(N)$, construct a field $\mathcal{K} = \text{Frac}(k[U]/(E_i))$, i.e., the fraction field of $\mathcal{R} = k[U]/(E_i)$. Let $\vec{f}_{\mathcal{K}} = \pi(\vec{f}) \in \mathcal{K}[X]^m$, where π is the canonical map. We claim that $\vec{f}_{\mathcal{K}}$ is unimodular over $\mathcal{K}[X]$. Then we can invoke the Quillen-Suslin Algorithm to complete $\vec{f}_{\mathcal{K}}$ to a unimodular matrix $P_{\mathcal{K}}$ over $\mathcal{K}[X]$.
- (4) Multiply the rows of $P_{\mathcal{K}}$ by some polynomial in $\mathcal{R}$ properly and obtain a new $P_{\mathcal{K}}$ such that $P_{\mathcal{K}} \in \mathcal{R}[X]^{m \times m}$. Suppose that $\det(P_{\mathcal{K}}) = d(U) \in \mathcal{R}$. Let $d_i = \pi^{-1}(d) \in k[U]$ and $P_i = \pi^{-1}(P_{\mathcal{K}})$ be a matrix over $k[U][X]$. Then we claim that:
- For any $\vec{a} \in V(E_i) \setminus (V(N) \cup V(d_i))$, $P_i(\vec{a})$ is unimodular and its first row is equal to $\vec{f}(\vec{a})$, i.e., $\vec{f}(\vec{a})$ is completed to the unimodular matrix $P_i(\vec{a})$ over $L[X]$.*
- (4) Furthermore, we consider $C'_i = V(E_i + \langle d_i \rangle) \setminus V(N)$ and repeat the processes (2)-(4) for C'_i if C'_i is not empty. Since $d \neq 0$ in $\mathcal{K}[X]$, then $d_i \notin E_i$, i.e., $E_i \subsetneq E_i + \langle d_i \rangle$. Therefore, this process will terminate within a finite number of steps.

In the terms of theory, we can complete the parametric unimodular row to unimodular matrix by the analysis above. However, in this process, we regard the Quillen-Suslin Algorithm as a black box and assume that we can invoke it to complete any unimodular row over $\mathcal{K}[X]$ for any field $\mathcal{K}$. Therefore, in terms of implementation, we need to consider how to implement Quillen-Suslin Algorithm over $\mathcal{K}[X]$ for any field $\mathcal{K}$, especially the field defined by a prime ideal as above, which is also called function field.

In summary, this paper extend the Quillen-Suslin Theorem to parametric case and present a process to complete a parametric row to unimodular matrix with parameters. However, in terms of implementation, further research is needed.

Keywords

Parametric unimodular rows, Quillen-Suslin Theorem, comprehensive Gröbner system.

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